SPACE POWER GENERATION UNITS

A. **Heat Engine (and turbogenerator)**; Like Rankine and Brayton cycles (regenerative)

B. Battery

1. Primary batteries consume their active materials and convert chemical into electrical energy. Not used since early 1960's.
2. Secondary batteries store electricity, utilize a reversible chemical reaction and can be recharged many times. Secondary batteries are used with solar cell systems for power augmentation, emergency back-up, and in 'shadowed' times when solar arrays are not functioning. Nickel-hydrogen batteries are used for backup charge on the International Space Station.

C. Fuel Cells

Chemical fuel cells are direct conversion devices practical for space power; duration 2 to 4 weeks and at power levels up to approx. 40 kW. Too heavy for long duration missions. 3 fuel cells in space shuttle have combined output of 21 kW at 27.5 Volts continuous or 36 kW for 15 seconds. H2 and O2 are conventional reactants. H2 gas is fed to an anode (-), O2 gas fed to a cathode (+); product is H2O. Reaction process also generates H+ ions and e- free electrons which establishes an external current from anode to cathode doing electrical work if an electrical load is attached. Heat given off must be removed to maintain temperature. High efficiency (approx. 50% possible).

D. Solar Cells

Solar cells rely on photovoltaic effect to directly convert solar radiation to electricity. Most long duration space missions to date use solar cells. Solar arrays have existed up to 70 kW (International Space Station). Solar cell arrays currently around 100 W/kg now with 250 W/kg possible (should be obtainable).
-Typical silicone solar cells have 20 to 35% efficiency, approx 20 Watts per square foot of array (?), maximum around 40%, maybe higher obtainable. Again, must convect away or re-radiate waste heat to maintain thermal equilibrium.
-Variations in temperature significantly affects electrical output of solar cells; may also need orientation machinery to turn arrays toward sunlight. Also need batteries, battery chargers, cooling systems, power conditioning units, electrical distribution system.
-Long life (> 10 years) possible.

E. Isotope Thermoelectric System

Energy provided is nuclear in origin, derived from decay of radioisotopes with long half-lives, energy is converted to heat and heat is (partly) converted to electricity by direct methods. Both U.S. and Russian spacecraft have used such systems.

A. Radioisotope thermoelectric generator; voltage created between two junctions of dissimilar metals at different temperatures.

B. Thermoionic power conversion cell; heat from isotope reactor raises temperature of a suitable material (like tungsten) to 1500 to 2000 degrees Centigrade so that it emits thermal electrons which are then captured by a cooled cathode.

-Both types require radiator to reject waste heat
-Advantage – long life, no moving parts, compact, simple
-Disadvantage – cost, low specific power, efficiency, possible contamination

SUMMARY OF POWER GENERATION

Batteries, fuel cells: low power, heavy, short life (2 kW and less, days)

Solar cells: low power, long life (10 kW and less), long life (years)

Solar cells, isotope thermoelectric: medium power, long life (10's kW and years)

Nuclear reactor, heat engine: high power, long life (100's kW to megaWatts, years)

POWER CONDITIONING UNITS

Power conditioning units needed to transform voltage and current due to mismatches between power generation units and thruster requirements. For example solar cells typically put out 28 to 110 Volts direct current. A small ion rocket needs 1000 to 2000 Volts or more…

-Power conditioning unit typically weighs more than the thruster.
- Power conditioning unit typically loses up to 20% of incoming power
-Has many parts (hundreds); very complex, services many various electrical needs, needs to be redundant
-If pulsed power in accelerator used, even more complex power conditioning required